

LI Kun

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**Director of Engineering, Perception - Deep
Learning - Autonomous Driving (Ref. ID:
JR1975848)**

Nvidia, Shanghai, China

Jan 28, 2026

Dear Hiring Manager,

I am writing to express my strong interest in the position of Director of Engineering, Perception in NVIDIA's Autonomous Driving Department. With over a decade of experience spanning AI, robotics, and full-stack autonomous systems, I am confident that I possess the strategic vision and technical leadership necessary to drive perception innovation within your organization.

Most recently, as Perception Department Head at Carizon — the joint venture between Volkswagen Group and Horizon Robotics — I led a team of 70 experts and was responsible for end-to-end perception delivery across multiple production vehicle programs. Under my leadership, we successfully delivered perception systems for several mass-production models with Anhui Volkswagen and SAIC Volkswagen, achieving key SOP milestones. A core achievement was building our in-house, unified data engine from the ground up — implementing a tightly-coupled, homogeneous data production pipeline and a closed-loop data system that continuously iterates on real-world driving scenarios. This infrastructure enabled the rapid development and deployment of our one-stage, multimodal BEV-fusion model, which has now been integrated into production vehicles.

Earlier in my career, I founded and scaled the perception team at Alibaba's Damo Academy. To drive commercialization, I later led the team's full transition into Cainiao Network, where we industrialized L4 unmanned delivery solutions and expanded the fleet to over a thousand vehicles in operational logistics. This provided me direct experience in scaling production-grade autonomous systems from research to widespread deployment.

I am deeply motivated by the challenge of building scalable, safety-critical perception systems and believe my background in both cutting-edge research and high-stakes product delivery would allow me to contribute meaningfully to NVIDIA's autonomous driving mission. I welcome the opportunity to discuss how my experience can support your team's goals.

Thank you for your time and consideration.

Sincerely yours,

LI Kun

Attached: curriculum vitae

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Perception Lead

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Education

2011-2015 **Doctor of Philosophy**, Department of Electrical and Computer Engineering, National University of Singapore (NUS), Singapore, CAP: 4.83/5.00.

2007-2011 **Bachelor in Control Science and Engineering**, College of Control Science and Engineering, Zhejiang University (ZJU), Hangzhou, China, CAP: 3.91/4.00.

Research Interest

AD & Robotics, End-to-End Models, Perception Fusion, BEV Perception, Generative Models, Vision/LiDAR Localization & SLAM, Fusion Algorithms.

Ph.D. thesis

Title *Navigation and Autonomous Control of MAVs in GPS-denied Environments.*
Supervisors Professor Ben M. Chen and Professor Tong H. Lee
Description My thesis developed a navigation and control system for a custom micro drone in GPS-denied environments. It fuses data from an IMU, optical flow sensor, and a monocular visual SLAM system for real-time positioning and 3D mapping.

Vocational Experience

2024.4 – **Perception Lead**, Carizon: A Joint-Venture Company by Volkswagen Group and
Present Horizon Robotics, Shanghai

Perception Department Head: Leading a team of 70. Responsible for the delivery of perception algorithms across vehicle product lines, including data ground-truth pipelines, onboard/cloud models, and perception fusion.

- **Ground-Truth Production Pipeline:** Established a pipeline based on continuous large-scale mapping, leveraging multimodal large models for multi-task pre-annotation (dynamic/static, occupancy/freespace). Enabled end-to-end homogeneous/heterogeneous data and synthetic data production.
- **Onboard Models:** Developed a one-stage multimodal BEV temporal fusion model (uni-model) for high-precision perception at 10fps. Researched next-generation VLM+E2E algorithms, designing end-to-end trajectory models via QWen 1B + Diffusion.
- **Cloud Models:** Utilized unified detection (G-Dino) and description (Qwen) models for efficient 4D/2D annotation. Employed 3DGS+Diffusion for editable scene reconstruction and controllable data synthesis.
- **Fusion Algorithms:** Developed modeling for static elements in mapless NOA solutions and implemented camera-radar fusion for dynamic obstacle tracking and state estimation.

2018.2 - **Staff Engineer in AD Lab**, *Damo Academy (Before 2023.5) & Cainiao Network*,
2024.4 *Alibaba*, Hang Zhou.

Perception TL Led a perception team of 50+ experts, responsible for perception, mapping, localization, and calibration across zero-occupant delivery vehicles (GX), intercity logistics vehicles (GT), and autonomous trucks (DX).

- **Multi-modality & Temporal Fusion Framework:** Built a unified BEV-based spatio-temporal fusion model for object detection, tracking, motion estimation, 2D/3D segmentation, occupancy prediction, freespace estimation, and lane/boundary detection with topology reasoning.
- **Advanced Visual Perception Systems:** Developed multi-camera fusion pipelines for detection, tracking, and state estimation, with modules for tiny-object detection, out-of-distribution segmentation, behavior recognition, driver monitoring (DMS), and traffic-light recognition & reasoning.
- **Real-Time Mapping & Localization Infrastructure:** Implemented tightly-coupled odometry fusing LiDAR, cameras, IMU, and encoders; engineered a real-time vector-map pipeline for element detection, topology compilation, and routing with standard-definition maps.
- **Offline Mapping & Automated 4D Labeling Platform:** Established an offline dense 4D reconstruction framework for static/dynamic scenes; led vector-map auto-generation with human-in-the-loop refinement and an end-to-end tile-centric crowd-sourced mapping system; introduced multi-task auto-labeling and proxy-task pre-training with unlabeled data.
- **Pre-Research Initiatives:** Explored multi-spectral and raw-camera (RCCB) visual models and pioneered an end-to-end multi-modal architecture integrating perception, prediction, and planning.

2017.4 - **Research Scientist in I²R**, *A*STAR*, Singapore.

2018.1 **Team Leader** of Autonomous Vehicle Localization Group, focusing on 3D map-based localization and mapping, system level software integration, and perception fusion with different sensor sources.

- **Monte-Carlo based Localization:** improvement on accuracy and high-speed capability of 2.5D monte carlo localization with multiple LiDAR sensors; recovery and re-localization from kidnapped situation and dynamic obstacle-strewn environment.
- **Onboard Integration:** ROS packages of multiple LiDAR sensors fusion, data calibration, distortion correction in high-speed scenario and time synchronization.

2015.9 - **Post-doctoral Research Fellow in Unmanned System Group**, *National University of Singapore*, Singapore.

Team Leader of the FSTD defense project "Onboard 3-Dimensional Navigation System for Unmanned Aerial Vehicles in Unknown and Realistic Indoor Environments", focusing on the SLAM algorithm, hardware system design and software modules development ([link](#)).

Detailed achievements:

- **Indoor Localization & Mapping:** a tightly-coupled visual-inertial odometry module fusing high-frequency inertial output and images to provide pose estimation and a local voxelized map simultaneously for trajectory planning.
- **Hardware Platform Development:** customized multi-sensory hexa-rotor UAV platform with GPS-denied environment navigation capability with LiDAR and stereo vision approaches.
- **GN&C Software Architecture:** multi-layer flight control firmware architecture with RTOS NuttX running on STM32F4 Cortex-M4 core as low-level avionic system; hierarchical modules (mission - planning - SLAM - reference) running on ROS with Intel Up-board.

Awards & Certificates

2023.11 **7th "Hangzhou Craftsman" in High-tech and AI fields**, *Hangzhou*, China

- 2023.6 **4th Place on 3D Object Detection Benchmark of Nuscenes Dataset.**
- 2022.9 **1st Place on Multiple Object Tracking Benchmark of MOT17.**
- 2022.7 **1st Place on Road Anomaly Segmentation & Road Obstacle Segmentation of SegmentMelfYouCan Dataset.**
- 2022.5 **1st Place on Road/Lane Detection of Kitti Dataset.**
- 2022.3 **1st Place on Depth Completion of Kitti Dataset.**
- 2021.6 **1st Place on Moving Object Segmentation (moving object detection with optical flow) of SemanticKitti Dataset.**
- 2017.10 **Best Paper Award at IEEE TENCON 2017 held in Penang, Malaysia.**
- 2016.2 **Multiple UAV light show on Future of Us at Gardens by the Bay (Sideshow), Singapore.**
- 2015.8 **2nd Prize in Outdoor Competition of Micro Air Vehicle Competition (IMAV 2015) held in Aachen, Germany, Germany.**
- 2014.8 **1st Place in the Micro Air Vehicle Competition (IMAV 2014) held in Delft, The Netherlands, The Netherlands.**
- 2014.8 **Presidential Scholarship by NUS, Singapore.**
- 2014.3 **1st Place in Singapore Amazing Flying Machine Competition (SAFMC) Category E, Singapore.**
- 2014.2 **Multiple UAV calligraphy demo on Singapore Airshow 2014 (Airshow), Singapore.**
- 2013.9 **New Innovation Star Award in International UAV Grand Prix (UAVGP 2013) held in Beijing, China.**
- 2013.2 **Best Presentation Reward (1st Place) In Control, Intelligent Systems and Robotics Track, NUS.**

Publications

- 2023 C Hu, H Zheng, **Kun Li**, et al, "*FusionFormer: A Multi-sensory Fusion in Bird's-Eye-View and Temporal Consistent Transformer for 3D Objection*", ICLR 2023.
- 2022 R Tian, Y Zhang, Z Cao, J Zhang, L Yang, S Coleman, D Kerr, **Kun Li**, "*Object SLAM With Robust Quadric Initialization and Mapping for Dynamic Outdoors*", IEEE ITS.
- 2021 **Kun Li**, Zhanpeng Ouyang, et al, "*Robust SRIF-based LiDAR-IMU Localization for Autonomous Vehicles*", IEEE ICRA, 2021.
- 2021 Yanjie Ke, **Kun Li**, et al, "*MDANet: Multi-Modal Deep Aggregation Network for Depth Completion*", IEEE ICRA, 2021.
- 2020 **Kun Li**, et al, "*Vision Global Localization with Semantic Segmentation and Interest Feature Points*", IEEE IROS, 2020.
- 2018 Jiaxin Li, Yingcai Bi, **Kun Li**, et al, "*Accurate 3d localization for mav swarms by uwb and imu fusion*", IEEE ICCA, 2018.
- 2017 Sai Prakhya, B. Liu, W. Lin, **K. Li**, et al, "*On creating low dimensional 3D feature descriptors with PCA*", TENCON, 2017.

- 2016 **K. Li**, et al, "*Improved efficiency electronic speed controller development for 3-Phase brushless DC motor in unmanned aerial systems*", IMAV, 2016.
- K. Li**, K. Wang, K. Zhang and B. M. Chen, "*Aggressive maneuvers of a quadrotor MAV based on composite nonlinear feedback control*", IEEE AIM, 2016.
- S. Lai, K. Wang, **K. Li** and B. M. Chen, "*Path planning of rotorcrafts in unknown environment*", CCC, 2016.
- K. Li**, K. Zhang and B. M. Chen, "*On-board visual odometry and autonomous control of a quadrotor micro aerial vehicle*", IEEE ICCA, 2016.
- 2015 **K. Li**, P. Liu, T. Pang, Z. Yang and B. M. Chen, "*Development of an unmanned aerial vehicle for rooftop landing and surveillance*", IEEE ICUAS, 2015.
- 2014 **K. Li**, R. Huang, et al, "*Off-board visual odometry and control of an ultralight quadrotor MAV*", IMAV, 2014.
- S. K. Phang, **K. Li**, et al, "*Systematic design and implementation of a micro unmanned quadrotor system*", Unmanned Systems, 2(2):1-21, 2014.
- S. K. Phang, **K. Li**, F. Wang, B. M. Chen and T. H. Lee, "*Explicit model identification and control of a micro aerial vehicle*", IEEE ICUAS, 2014.
- 2013 **K. Li**, S. K. Phang, B. M. Chen and T. H. Lee, "*Platform design and mathematical modeling of an ultralight quadrotor micro aerial vehicle*", IEEE ICUAS, 2013.
- S. K. Phang, **K. Li**, "*Systematic design methodology and construction of micro aerial quadrotor vehicles*", Handbook of Unmanned Aerial Vehicles, 2013.